# QUANTUM TECHNOLOGIES AS STRATEGIC INFRASTRUCTURE: EVIDENCE, ADOPTION PATHWAYS, AND A NATIONAL RESEARCH AGENDA (2025-2035)

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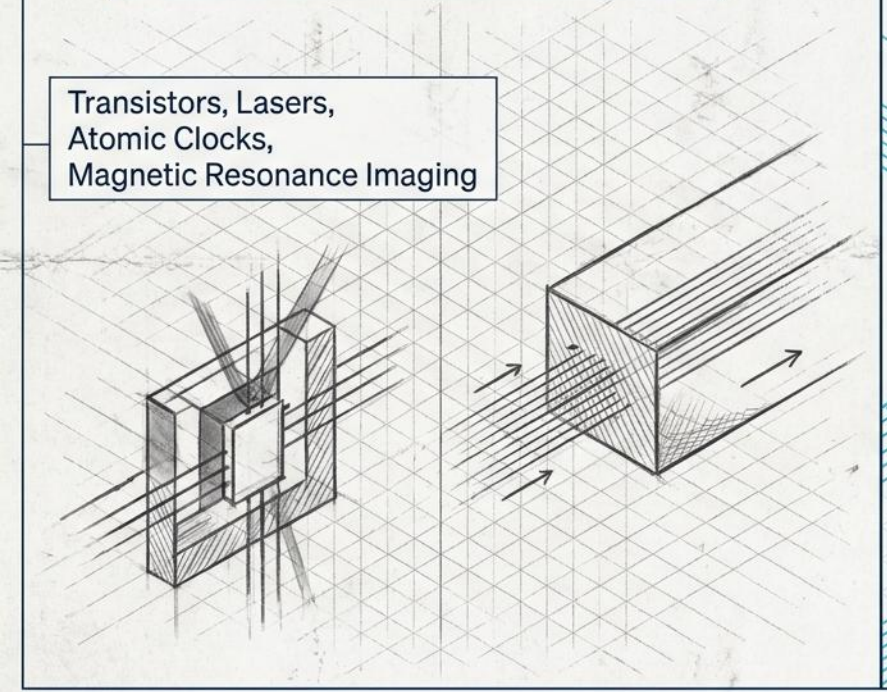
Quantum technology is transitioning from microscopic physics to macro-scale engineered infrastructure.

## The First Quantum Revolution

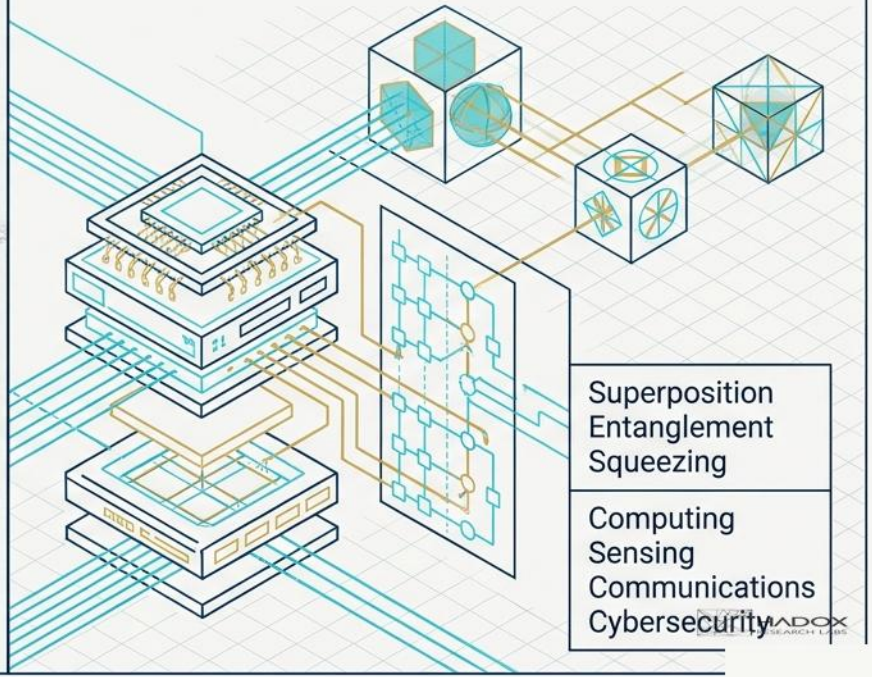
Bulk Properties & Statistical Phenomena

Transistors, Lasers,
Atomic Clocks,
Magnetic Resonance Imaging



## The Second Quantum Revolution

Direct Engineering of Quantum States



Superposition
Entanglement
Squeezing

Computing
Sensing
Communications
Cybersecurity

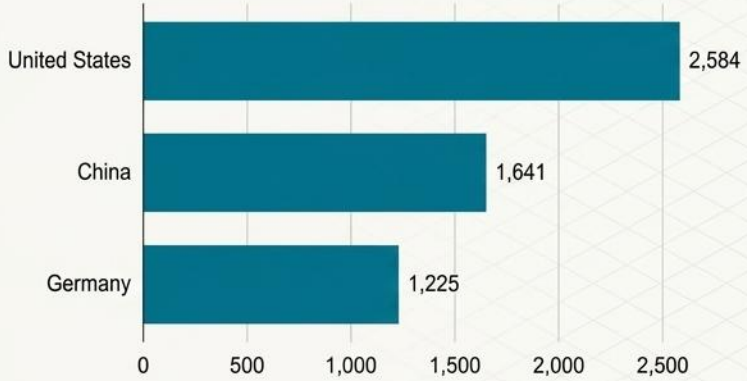
## The empirical record reveals massive acceleration alongside structural and geographic unevenness.

**$55.7B**
Public Commitments worldwide (Oct 2025)

**$11.235B**
PPP in 12,209 public R&D awards (2015-2023)

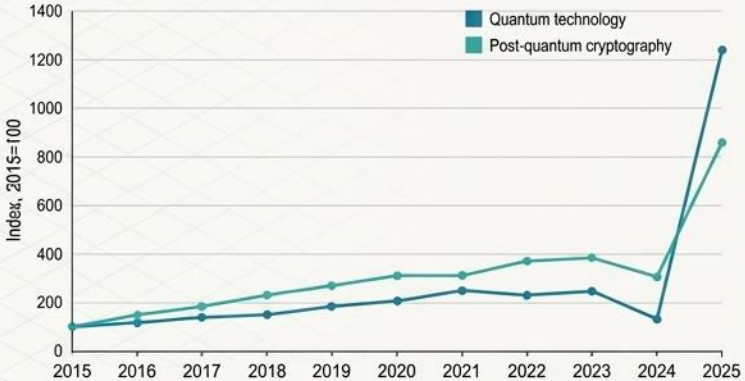
**31,700**
Quantum patent families (2005-2024)

### Geographic Concentration of Scientific Capacity




Source: OpenAlex Works API extraction. Counts are bibliometric adoption proxies, not direct deployment measures.

### Exponential Expansion of Literature




Source: OpenAlex Works API extraction. The 2025 quantum-technology spike is interpreted as an indexing/query signal requiring cautious interpretation.

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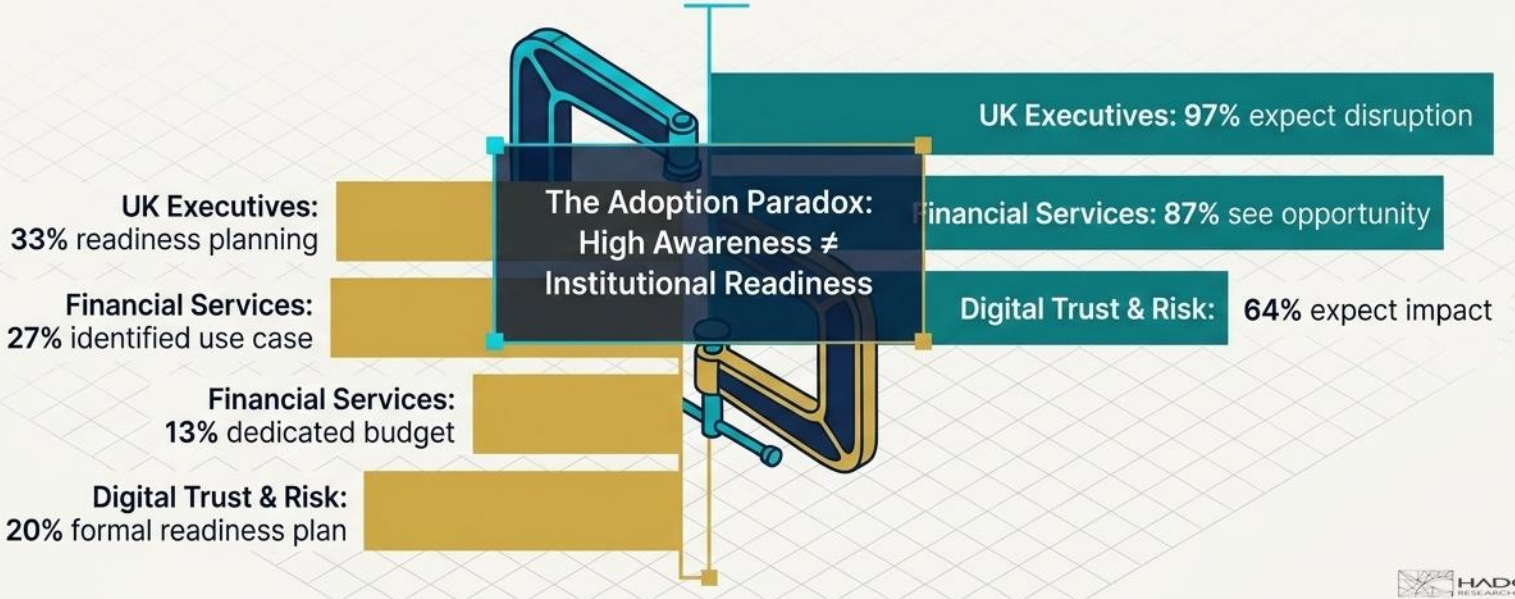
Editorial Blueprint

# An extreme awareness-readiness gap currently paralyzes enterprise adoption.

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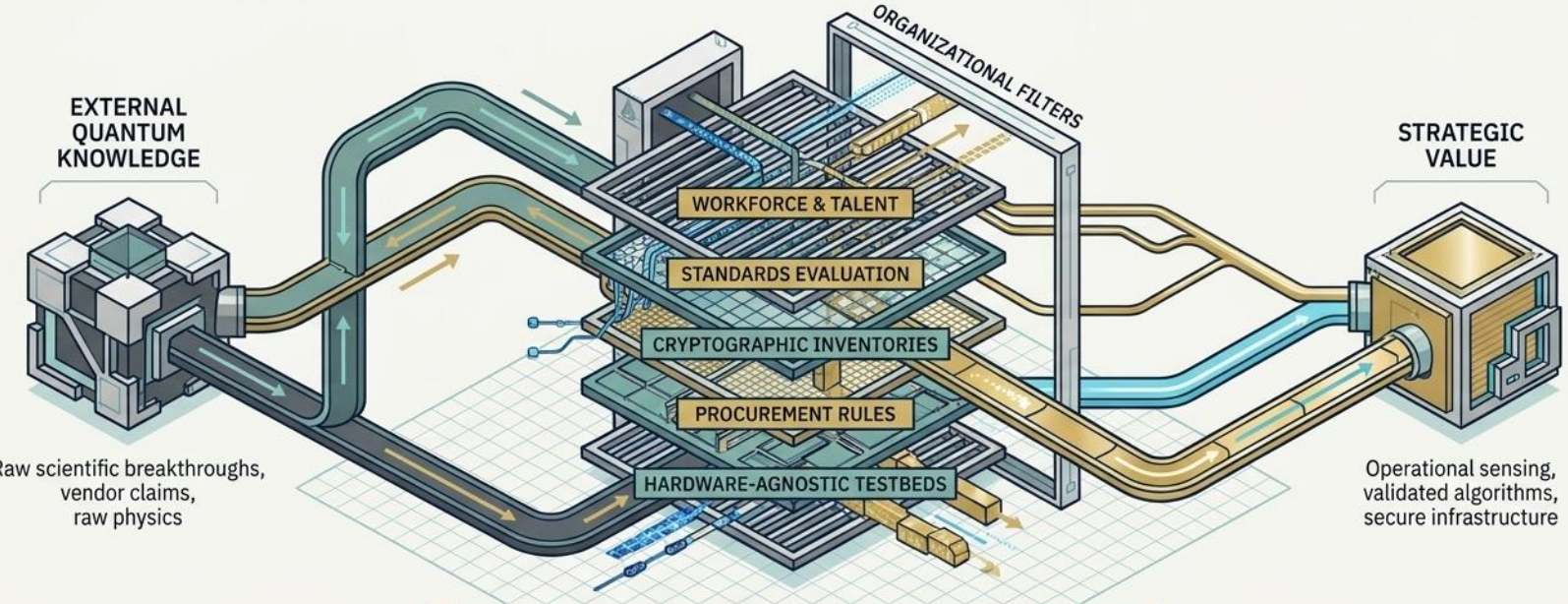
# Quantum adoption is an absorptive capacity problem, not simply a hardware problem.

**Absorptive Capacity** (Cohen & Levinthal, 1990): The ability to recognize, assimilate, and exploit external knowledge.




**RAW HARDWARE CAPABILITY CANNOT BYPASS ORGANIZATIONAL FILTERS.**

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# The quantum ecosystem forms a layered capability stack with distinct adoption timelines

Low Infrastructure Burden | High Infrastructure Burden

High Maturity

**Post-Quantum Cryptography (PQC)**
Immediate Security Mandate

**Quantum Sensing**
Near-Term Public & Industrial Value

Emerging Maturity

**Quantum Computing**
High Optionality, Low Near-Term Certainty

**Quantum Communication**
Specialized Infrastructure

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(Data Shelf Life + Migration Time) > Time to Quantum Computer = System Vulnerable Today

**Harvest Now, Decrypt Later**
Data collection happens in the present.

**NIST 2024 Standards Finalized**
ML-KEM, ML-DSA, SLH-DSA

**Time to Quantum Computer**

**Post-Quantum Cryptography (PQC) is the first undeniable digital-infrastructure transition.**
PQC requires multi-year cryptographic inventories, crypto-agility, and legacy system replacement.

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# Strategic evaluation requires shifting metrics from physical qubits to logical-qubit economics

**Metric Shift:**
Physical Qubits → Logical-Qubit Economics

| | Strength | Bottleneck | Strategic View |
|---|---|---|---|
| Superconducting | Fast gates, industrial ecosystem | Cryogenic wiring, thermal load | Scale depends on error correction |
| Trapped Ions | High fidelity, long coherence | Gate speed, shuttling | Engineering scale/throughput remain central |
| Neutral Atoms | Large arrays, flexible geometry | Atom loss, gate fidelity | Strong option for simulation |
| Photonics | Network compatibility, room-temp | Photon loss, deterministic gates | Crucial for networking |
| Annealing | Available systems | Limited universality | Must be benchmarked against classical heuristics |

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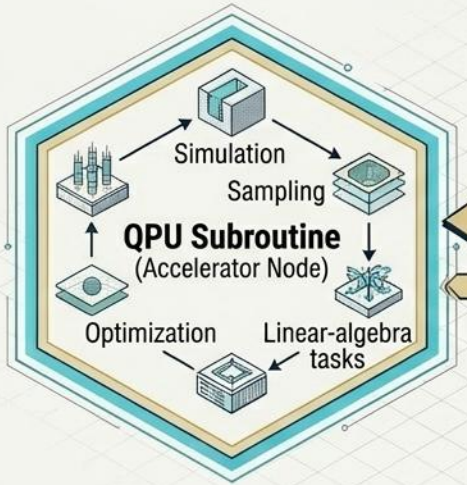
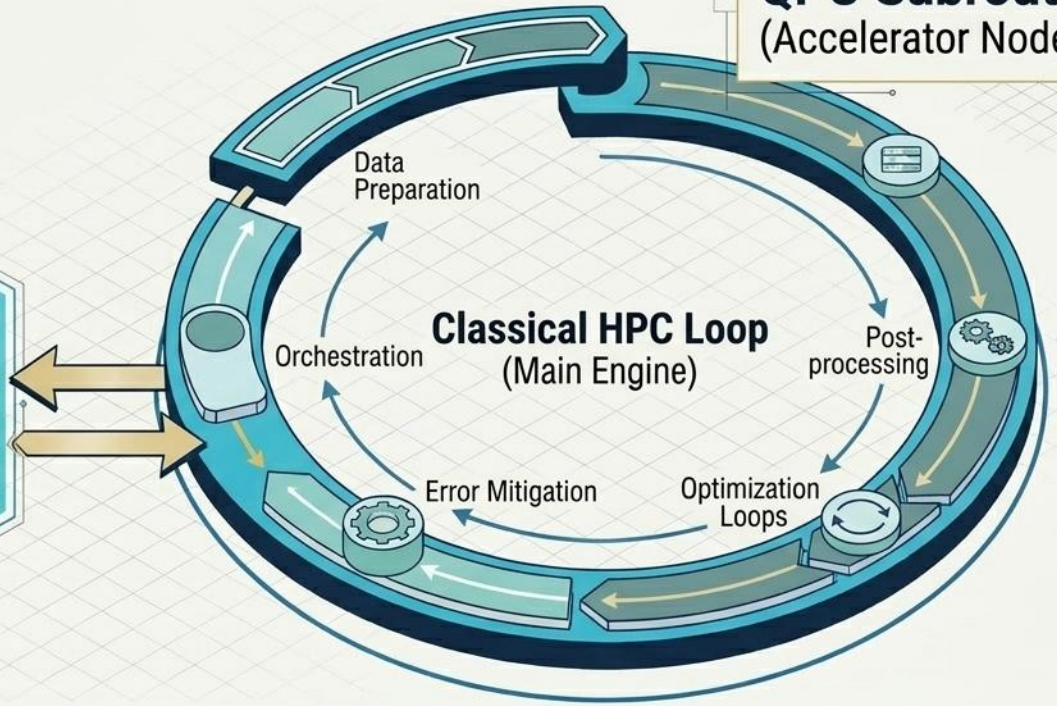
# Architectural Workflow




The strategic objective is an HPC-QC continuum where supercomputing remains the operational substrate, with QPUs invoked under realistic latency, noise, and verification constraints.

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# Scientific relevance does not automatically guarantee enterprise or industrial adoption.

## Chemistry & Materials

Strongest Long-Run Case

Simulates quantum systems naturally, but requires hybrid algorithms and integration with lab workflows.

## Optimization

High Hype, Difficult Baselines

Must beat exceptionally strong classical mixed-integer programming and GPUs. Currently exploratory.

## Finance

Demanding Viability

Must improve speed, regulatory auditability, and model quality compared to classical Monte Carlo.

## Machine Learning & AI

Research Stage

Suffers from barren plateaus. AI is more likely to improve quantum systems than vice versa right now.

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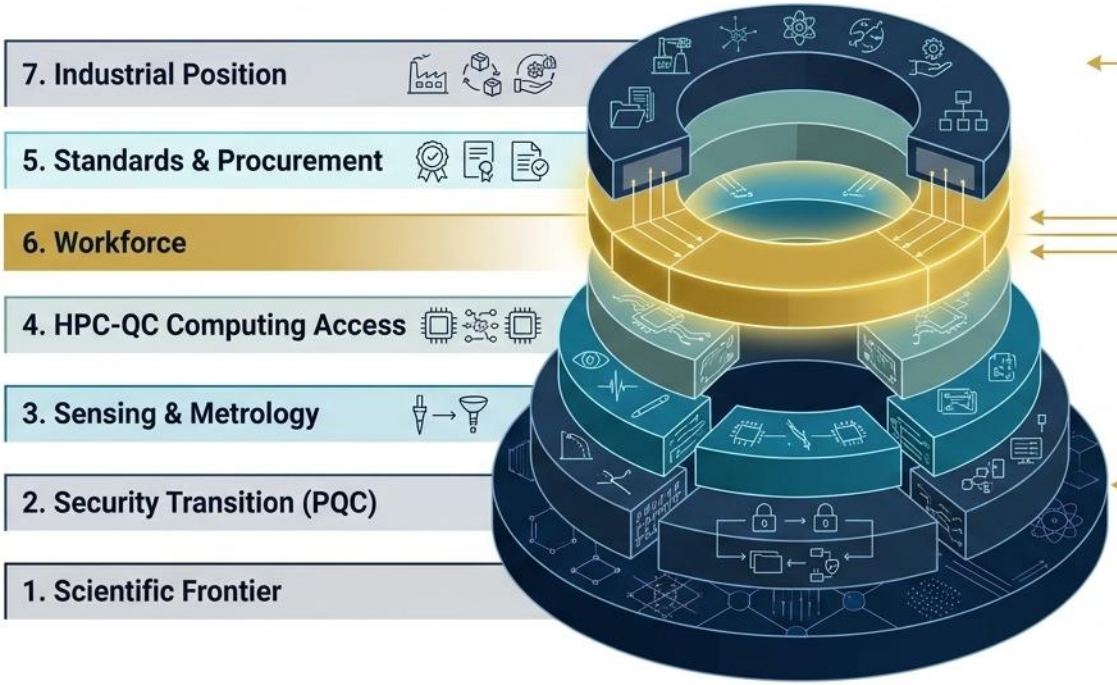
# A seven-pillar capability model separates operational readiness from rhetorical innovation policy




**The Missing Middle**

A nation cannot rely only on physics PhDs; it needs technicians, control system engineers, cybersecurity auditors, and procurement officers to translate science into infrastructure.

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# National Quantum Roadmap and Strategic Capability Sequencing, 2025-2035

Capability accumulation proceeds through staged preparation, operational testbeds, and selective scaling. Roadmap phases represent capability milestones, not hardware-breakthrough predictions.

**2025-2027 Preparation & Security**

- PQC inventories, QCaaS access, workforce curricula, sensing feasibility

**2027-2030 Testbeds & Pilots**

- Sensing pilots, PQC staged migration, benchmark labs, justified QKD

**2030-2035 Selective Scaling**

- Operational sensing, quantum networks, logical-qubit readiness, frontier research

| | 2025-2027 Preparation | 2027-2030 Testbeds | 2030-2035 Scaling |
|---|---|---|---|
| PQC | 3 | 3 | 3 |
| Sensing | 2 | 3 | 3 |
| QCaaS | 2 | 3 | 2 |
| Communication | 1 | 2 | 3 |
| Standards | 2 | 3 | 3 |
| Workforce | 3 | 3 | 3 |
| Frontier R&D | 2 | 2 | 3 |

not central | watch | build | priority

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# Six strategic errors consistently threaten enterprise decision-makers and policy scholars.

## 1. Waiting

Delaying action until fault-tolerant machines arrive ignores urgent PQC and sensing mandates.

## 2. Hardware Nationalism

Buying a domestic computer without a trained workforce creates a symbolic asset, not a capability.

## 3. Confusing QKD with PQC

Treating Quantum Key Distribution as a broad substitute for PQC misreads the infrastructure problem.

## 4. Treating QCaaS as Neutral

Cloud access democratizes experimentation but concentrates platform power and data governance.

## 5. Baselineless Advantage Claims

Accepting speedup claims without specifying the classical baseline, error model, and economic relevance.

## 6. Underfunding the Missing Middle

Ignoring the technicians and engineers needed to operate the systems.

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Quantum readiness is a cultivated institutional capability, not a hardware purchase.

# Quantum strategy succeeds when it turns fragile quantum effects into reliable public capability.

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# Evidence base and source repository


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**Repository:** github.com/ekaropolus/quantum-technologies-national-strategy
**Main article:** quantum_technologies_research_paper.pdf
**Slide map:** reveal_quantum_strategic_infrastructure/paper_slide_map.md

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